

LECTURE NOTES ON INTEGRABLE SYSTEMS

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ABSTRACT. These notes are based on the course *Introduction to Integrable Systems*, delivered by Professor Liming Ling during the Fall semester of 2025–2026.

The full L^AT_EX source code of these notes is available at: <https://codeberg.org/kill11kill/courseNotes>.

Week 1: Algebro-Geometric Solutions for the mKdV Equation (1)

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Objectives: Derive the mKdV equation from the associated spectral problem and construct the AKNS hierarchy by explicit recursion.

The focusing modified Korteweg–de Vries (mKdV) equation is given by

$$(1.1) \quad u_t + 6u^2u_x + u_{xxx} = 0, \quad u(x, t) \in \mathbb{R}, \quad (x, t) \in \mathbb{R}^2.$$

A standard method for establishing integrability is via a Lax pair. We introduce a matrix function $\Phi(x, t)$ satisfying

$$\Phi_x = U\Phi, \quad \Phi_t = V\Phi,$$

where U and V are 2×2 matrices depending on u and its derivatives. The compatibility condition

$$\Phi_{xt} = \Phi_{tx} \iff U_t - V_x + [U, V] = 0,$$

with $[U, V] = UV - VU$, is equivalent to the zero-curvature equation.

Following the AKNS formalism, the spatial part of the Lax pair is taken as

$$(1.2) \quad \Phi_x = U\Phi, \quad U = -i\lambda\sigma_3 + Q,$$

where

$$(1.3) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad \sigma_3 = \text{diag}(1, -1).$$

We introduce the transformation

$$(1.4) \quad \Phi(x; \lambda) = m(x; \lambda)e^{-i\lambda x\sigma_3}, \quad m(x; \lambda) \rightarrow \mathbb{I} \text{ as } \lambda \rightarrow \infty.$$

Substituting into the spectral problem (1.2) gives

$$(1.5) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i\lambda \sigma_3 + Q.$$

Define

$$(1.6) \quad \Psi = m \sigma_3 m^{-1}.$$

Differentiating Ψ and using (1.5), we compute

$$\begin{aligned} \Psi_x &= m_x \sigma_3 m^{-1} - m \sigma_3 m^{-1} m_x m^{-1} \\ &= [m_x m^{-1}, m \sigma_3 m^{-1}] \\ &= [m_x m^{-1} - i\lambda m \sigma_3 m^{-1}, m \sigma_3 m^{-1}] \\ &= [U, \Psi]. \end{aligned}$$

Thus Ψ satisfies

$$(1.7) \quad \Psi_x = [U, \Psi],$$

which is the stationary zero-curvature equation.

The matrix $m(x; \lambda)$ admits the asymptotic expansion

$$(1.8) \quad m(x; \lambda) = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + \dots.$$

From (1.6) and (1.8) we obtain the formal expansion

$$(1.9) \quad \Psi = \Psi_0 + \Psi_1\lambda^{-1} + \Psi_2\lambda^{-2} + \dots, \quad \Psi_0 = \sigma_3.$$

Substituting this into (1.7) yields

$$\begin{aligned} \sum_{j=1}^{\infty} \Psi_{j,x} \lambda^{-j} &= -i\lambda \left[\sigma_3, \sum_{j=1}^{\infty} \Psi_j \lambda^{-j} \right] + \left[Q, \sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right] \\ &= -i \sum_{j=0}^{\infty} \left[\sigma_3, \Psi_{j+1} \right] \lambda^{-j} + \sum_{j=0}^{\infty} \left[Q, \Psi_j \right] \lambda^{-j}. \end{aligned}$$

At order $O(1)$ we find

$$-i[\sigma_3, \Psi_1] + [Q, \sigma_3] = 0 \iff [i\Psi_1 + Q, \sigma_3] = 0,$$

which implies

$$\Psi_1^{\text{off}} = iQ.$$

For general $j \geq 1$, the coefficient of λ^{-j} gives

$$(1.10) \quad \Psi_{j,x} = -i[\sigma_3, \Psi_{j+1}] + [Q, \Psi_j] = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j].$$

Hence

$$(1.11) \quad \Psi_{j,x}^{\text{off}} = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j^{\text{diag}}] \iff \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}]),$$

and

$$(1.12) \quad \Psi_{j,x}^{\text{diag}} = [Q, \Psi_j^{\text{off}}] \iff \Psi_j^{\text{diag}} = \partial_x^{-1}([Q, \Psi_j^{\text{off}}]).$$

Thus, the recursion relation (1.11) provides a direct way to compute the off-diagonal coefficients algebraically in terms of lower-order terms, while the diagonal recursion (1.12) requires performing an integration. In practice, this makes the off-diagonal part more convenient to obtain explicitly, whereas the diagonal part is generally harder to compute in closed form.

By (1.6), the additional constraint $\Psi^2 = \mathbb{I}$ provides algebraic relations among the coefficients. Substituting the expansion (1.9) yields

$$\mathbb{I} = \left(\sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right)^2 = \sum_{j=0}^{\infty} \left(\sum_{k=0}^j \Psi_k \Psi_{j-k} \right) \lambda^{-j},$$

so that

$$(1.13) \quad \sum_{k=0}^j \Psi_k \Psi_{j-k} = 0, \quad j \geq 1.$$

Separating diagonal and off-diagonal parts, for $j \geq 1$ we obtain

$$(1.14) \quad \sum_{k=0}^j (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}) = 0,$$

$$(1.15) \quad \sum_{k=0}^j (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Since $\Psi_0 = \sigma_3$, the off-diagonal constraint (1.15) for $j \geq 2$ takes the form

$$\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 + \sum_{k=1}^{j-1} (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Noting that $\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 = 0$, this simplifies to

$$(1.16) \quad \sum_{k=1}^{j-1} \left(\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}} \right) = 0.$$

It should be noted that from the off-diagonal constraint (1.15) one can only deduce the reduced form (1.16), which merely expresses a compatibility condition between lower-order coefficients. In particular, this relation does not provide a practical means to compute the diagonal parts explicitly.

For the diagonal part (1.14) we have

$$\sigma_3 \Psi_j^{\text{diag}} + \Psi_j^{\text{diag}} \sigma_3 + \sum_{k=1}^{j-1} \left(\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right) = 0, \quad j \geq 2.$$

Since $\sigma_3 \Psi_j^{\text{diag}} = \Psi_j^{\text{diag}} \sigma_3$, this simplifies to

$$(1.17) \quad \Psi_j^{\text{diag}} = -\frac{1}{2} \sigma_3 \sum_{k=1}^{j-1} \left(\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right), \quad j \geq 2.$$

We will use this reduced relation (1.17) to compute the diagonal part Ψ_j^{diag} recursively at each order. This provides an explicit algebraic formula for the diagonal coefficients in terms of lower-order diagonal and off-diagonal terms.

For $j = 1$, the condition (1.13) reads

$$\sigma_3 \Psi_1 + \Psi_1 \sigma_3 = 2\sigma_3 \Psi_1^{\text{diag}} = 0,$$

so that $\Psi_1 = \Psi_1^{\text{off}} = iQ$.

Using (1.11) and (1.17) recursively, we find

$$\Psi_2^{\text{off}} = -\frac{1}{2} \sigma_3 Q_x, \quad \Psi_2^{\text{diag}} = \frac{1}{2} \sigma_3 Q^2,$$

and hence

$$\Psi_2 = \frac{1}{2} \sigma_3 (Q^2 - Q_x).$$

For the next coefficient, we compute

$$\begin{aligned} \Psi_3^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{2,x}^{\text{off}} - [Q, \Psi_2^{\text{diag}}]) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - [Q, \frac{1}{2} \sigma_3 Q^2] \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - \frac{1}{2} (Q \sigma_3 Q^2 - \sigma_3 Q^2 Q) \right) \\ &= -\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3. \end{aligned}$$

For the diagonal part,

$$\begin{aligned} \Psi_3^{\text{diag}} &= -\frac{1}{2} \sigma_3 \left(\Psi_1^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{off}} \Psi_1^{\text{off}} + \Psi_1^{\text{diag}} \Psi_2^{\text{diag}} + \Psi_2^{\text{diag}} \Psi_1^{\text{diag}} \right) \\ &= -\frac{1}{2} \sigma_3 \left((iQ) \left(-\frac{1}{2} \sigma_3 Q_x \right) + \left(-\frac{1}{2} \sigma_3 Q_x \right) (iQ) \right) \\ &= -\frac{1}{2} \sigma_3 \left(-\frac{i}{2} (Q \sigma_3 Q_x + \sigma_3 Q_x Q) \right) \\ &= \frac{i}{4} (Q_x Q - Q Q_x) = 0. \end{aligned}$$

Proceeding to Ψ_4 , the off-diagonal recursion (1.11) with $\Psi_3^{\text{diag}} = 0$ gives

$$\begin{aligned} \Psi_4^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{3,x}^{\text{off}} - [Q, \Psi_3^{\text{diag}}]) = \frac{i}{2} \sigma_3 \Psi_{3,x}^{\text{off}} \\ &= \frac{i}{2} \sigma_3 \partial_x \left(-\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3 \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{i}{4} Q_{xxx} + \frac{i}{2} (Q^3)_x \right) \\ &= \frac{1}{8} \sigma_3 Q_{xxx} - \frac{1}{4} \sigma_3 (Q^3)_x. \end{aligned}$$

Since $Q_x Q = Q Q_x$ for the present Q given in (1.3), we have

$$(Q^3)_x = Q_x Q^2 + Q Q_x Q + Q^2 Q_x = 3Q_x Q^2,$$

and therefore

$$\Psi_4^{\text{off}} = \frac{1}{8}\sigma_3 Q_{xxx} - \frac{3}{4}\sigma_3 Q_x Q^2.$$

For the diagonal part, only the nonvanishing combinations contribute; terms involving Ψ_1^{diag} or Ψ_3^{diag} are zero and omitted. Hence

$$\begin{aligned} \Psi_4^{\text{diag}} &= -\frac{1}{2}\sigma_3 \left(\Psi_1^{\text{off}} \Psi_3^{\text{off}} + \Psi_3^{\text{off}} \Psi_1^{\text{off}} + \Psi_2^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{diag}} \Psi_2^{\text{diag}} \right) \\ &= -\frac{1}{2}\sigma_3 \left((iQ) \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) + \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) (iQ) \right. \\ &\quad \left. + \left(-\frac{1}{2}\sigma_3 Q_x \right)^2 + \left(\frac{1}{2}\sigma_3 Q^2 \right)^2 \right) \\ &= -\frac{1}{8}\sigma_3 (Q Q_{xx} + Q_{xx} Q - 3Q^4 - Q_x^2). \end{aligned}$$

Since $(-i\lambda\Psi)_+ = -i\lambda\sigma_3 + Q$, we obtain

$$\Phi_x \Phi^{-1} = U = (-i\lambda\Psi)_+.$$

Here the subscript $(\cdot)_+$ denotes the projection onto the nonnegative powers of λ in the Laurent expansion. This motivates the definition of the AKNS hierarchy by prescribing the time evolution

$$(1.18) \quad \Phi_{t_n} = (-i\lambda^n \Psi)_+ \Phi \equiv V_n \Phi.$$

The first few members of the hierarchy are given by

$$\begin{aligned} n = 2 &\Rightarrow \text{nonlinear Schrödinger (NLS) flow,} \\ n = 3 &\Rightarrow \text{complex mKdV flow,} \Rightarrow \text{mKdV flow.} \end{aligned}$$

In terms of the expansion coefficients Ψ_j , the operators V_n take the form

$$\begin{aligned} V_2 &= -i(\lambda^2 \Psi_0 + \lambda \Psi_1 + \Psi_2), \\ V_3 &= -i(\lambda^3 \Psi_0 + \lambda^2 \Psi_1 + \lambda \Psi_2 + \Psi_3). \end{aligned}$$

The zero-curvature equation $U_t - V_x + [U, V] = 0$ with $V_{\text{mKdV}} = 4V_3$, yields at order $O(1)$

$$Q_t + 4i\Psi_{3,x} + [Q, -4i\Psi_3] = 0.$$

Taking the off-diagonal part gives

$$Q_t + 4i\Psi_{3,x}^{\text{off}} = 0,$$

that is,

$$(1.19) \quad Q_t + 4i \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right)_x = Q_t + Q_{xxx} - 2(Q^3)_x = 0.$$

Since

$$Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad Q^2 = -u^2 \mathbb{I},$$

this reduces to the scalar equation

$$u_t + u_{xxx} + 2(u^3)_x = 0,$$

which is precisely the mKdV equation (1.1).

Supplementary computations:

Using the definitions (1.4) and (1.6) we have

$$(1.20) \quad \Psi = \Phi \sigma_3 \Phi^{-1}.$$

Differentiating Ψ with respect to t_n and using (1.18) gives

$$\begin{aligned} \Psi_{t_n} &= \Phi_{t_n} \sigma_3 \Phi^{-1} - \Phi \sigma_3 \Phi^{-1} \Phi_{t_n} \Phi^{-1} \\ &= [\Phi_{t_n} \Phi^{-1}, \Psi] = [V_n, \Psi]. \end{aligned}$$

Thus

$$(1.21) \quad \Psi_{t_n} = [V_n, \Psi].$$

Next we simplify the zero-curvature equation for the pair (U, V_n) . The zero-curvature equation is

$$U_{t_n} - (V_n)_x + [U, V_n] = 0,$$

and since $U = -i\lambda\sigma_3 + Q$ we have $U_{t_n} = Q_{t_n}$. With the normalization

$$V_n = \alpha_n (-i\lambda^n \Psi)_+$$

we obtain

$$Q_{t_n} = (V_n)_x - [U, V_n] = \alpha_n (-i\lambda^n \Psi_x)_+ - [U, \alpha_n (-i\lambda^n \Psi)_+].$$

Pulling out constants and using $\Psi_x = [U, \Psi]$ gives

$$(1.22) \quad Q_{t_n} = -i\alpha_n \left((\lambda^n [U, \Psi])_+ - [U, (\lambda^n \Psi)_+] \right).$$

Write $U = -i\lambda\sigma_3 + Q$ and split the first term in (1.22):

$$\lambda^n [U, \Psi] = -i\lambda^{n+1} [\sigma_3, \Psi] + \lambda^n [Q, \Psi].$$

Because Q does not depend on λ we have

$$(\lambda^n [Q, \Psi])_+ = [Q, (\lambda^n \Psi)_+],$$

so the Q -commutator contributions cancel in (1.22). The remaining terms therefore come only from the σ_3 -commutator:

$$\begin{aligned} Q_{t_n} &= -i\alpha_n \left(-i(\lambda^{n+1} [\sigma_3, \Psi])_+ + i\lambda [\sigma_3, (\lambda^n \Psi)_+] \right) \\ &= \alpha_n \left(\lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ \right). \end{aligned}$$

To simplify the difference of finite sums, use the expansions

$$\Psi = \sum_{j \geq 0} \Psi_j \lambda^{-j}, \quad (\lambda^n \Psi)_+ = \sum_{j=0}^n \lambda^{n-j} \Psi_j.$$

A direct comparison of coefficients shows

$$\begin{aligned} \lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ &= \sum_{j=0}^n \lambda^{n+1-j} [\sigma_3, \Psi_j] - \sum_{j=0}^{n+1} \lambda^{n+1-j} [\sigma_3, \Psi_j] \\ &= -\lambda^{n+1-(n+1)} [\sigma_3, \Psi_{n+1}] \\ &= -[\sigma_3, \Psi_{n+1}]. \end{aligned}$$

Hence the evolution of Q is

$$(1.23) \quad Q_{t_n} = -\alpha_n [\sigma_3, \Psi_{n+1}].$$

Since diagonal parts commute with σ_3 and the commutator doubles the off-diagonal part, we may equivalently write

$$(1.24) \quad Q_{t_n} = -2\alpha_n \sigma_3 \Psi_{n+1}^{\text{off}}.$$

Objectives: Construct the L matrix for mKdV, establish its symmetries and conserved determinant, and relate the eigen-data to the associated hyperelliptic curve and Riemann–Theta functions.

Recall the AKNS spectral problem (1974)

$$(2.1) \quad \Phi_x = U(\lambda; Q) \Phi, \quad U(\lambda; Q) = -i\lambda\sigma_3 + Q,$$

where $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and Q is off-diagonal. In what follows we set

$$(2.2) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}.$$

Introduce the gauge

$$(2.3) \quad \Phi = m(x; \lambda) e^{-i\lambda x \sigma_3}, \quad m = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + O(\lambda^{-3}),$$

$$(2.4) \quad \Psi := m \sigma_3 m^{-1} = \Psi_0 + \Psi_1 \lambda^{-1} + \Psi_2 \lambda^{-2} + \dots,$$

so that $\Psi^2 = \mathbb{I}$ and Ψ satisfies the stationary zero-curvature equation

$$(2.5) \quad \Psi_x = [U, \Psi].$$

Decompose a matrix A into diagonal and off-diagonal parts $A = A^{\text{diag}} + A^{\text{off}}$. From (2.5) and $\Psi^2 = \mathbb{I}$ one obtains the recursion

$$(2.6) \quad \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3 \left(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}] \right), \quad j \geq 0,$$

$$(2.7) \quad \Psi_{j+1}^{\text{diag}} = -\frac{1}{2}\sigma_3 \sum_{k=1}^j \left(\Psi_k^{\text{diag}} \Psi_{j-k+1}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k+1}^{\text{off}} \right), \quad j \geq 1.$$

Hence each Ψ_j is a differential polynomial in Q and its x -derivatives.

One checks that

$$(2.8) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i(\lambda \Psi)_+.$$

Let $\Phi(x, t_n; \lambda) = m(x, t_n; \lambda) e^{-i(\lambda x + \lambda^n t_n) \sigma_3}$, then

$$(2.9) \quad \Phi_{t_n} \Phi^{-1} = m_{t_n} m^{-1} - i\lambda^n m \sigma_3 m^{-1} = -i(\lambda^n \Psi)_+ \equiv V_n,$$

so that $V_n = -i(\lambda^n \Psi)_+$ and

$$(2.10) \quad \Psi_{t_n} = [V_n, \Psi].$$

The compatibility $\Phi_{xt_n} = \Phi_{t_n x}$ is equivalent to the zero curvature equation

$$(2.11) \quad U_{t_n} - V_{n,x} + [U, V_n] = 0.$$

Definition 1. Let $g \geq 0$ be a fixed integer, and let $\{\alpha_j\}_{j=0}^{g+1}$ be constants. Define

$$(2.12) \quad L := -i \sum_{j=0}^{g+1} \alpha_j (\lambda^j \Psi(x, t, \lambda))_+,$$

where $(\cdot)_+$ denotes the projection onto nonnegative powers of λ .

It is convenient to rewrite L by expanding the nonnegative part explicitly. Using $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$ we have

$$(\lambda^j \Psi)_+ = \sum_{k=0}^j \Psi_k \lambda^{j-k},$$

hence

$$(2.13) \quad L(\lambda) = -i \sum_{j=0}^{g+1} \alpha_j \sum_{k=0}^j \Psi_k \lambda^{j-k} = -i \sum_{l=0}^{g+1} \left(\sum_{j=l}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set $l = j - k$ to obtain the last form.

Proposition 1. Suppose L as defined in (2.12) satisfies the stationary zero-curvature equations

$$(2.14) \quad L_x = [U, L], \quad L_{t_n} = [V_n, L].$$

Then the following additional constraint holds:

$$(2.15) \quad \sum_{j=1}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}}(x, t_n) + i \alpha_0 Q = 0.$$

Proof. Set

$$(2.16) \quad \widehat{L}(\lambda) := -i \left(\sum_{j=0}^{g+1} \alpha_j \lambda^j \right) \Psi(\lambda).$$

By construction $\widehat{L}_x = [U, \widehat{L}]$. Expand $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$ and substitute into $\widehat{L}$:

$$(2.17) \quad \widehat{L}(\lambda) = -i \sum_{j=0}^{g+1} \sum_{k \geq 0} \alpha_j \Psi_k \lambda^{j-k} = -i \sum_{l=-\infty}^{g+1} \left(\sum_{j=0}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set $l = j - k$. Decompose $\widehat{L} = \widehat{L}_+ + \widehat{L}_-$ into nonnegative and negative powers of λ . From $\widehat{L}_x = [U, \widehat{L}]$ we obtain

$$\left(\widehat{L}_+ + \widehat{L}_- \right)_x = [U, \widehat{L}_+ + \widehat{L}_-].$$

After extracting nonnegative powers of λ , we have

$$(\widehat{L}_+)_x - [U, \widehat{L}_+] = \left([-i\lambda\sigma_3, \widehat{L}_-] \right)_+ = -i \left[\sigma_3, \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1} \right] = -2i\sigma_3 \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}} = 0,$$

which yields the constraint (2.15).

Next, to ensure compatibility, we require to prove $L_{xt_n} = L_{t_n x}$. Observe that

$$\begin{aligned} L_{xt_n} &= [U_{t_n}, L] + [U, [V_n, L]], \\ L_{t_n x} &= [V_{n,x}, L] + [V_n, [U, L]]. \end{aligned}$$

Applying the Jacobi identity,

$$[[A, B], C] + [[C, A], B] + [[B, C], A] = 0,$$

we find

$$[U, [V_n, L]] - [V_n, [U, L]] = [[L, V_n], U] + [[U, L], V_n] = [[U, V_n], L].$$

Therefore, it follows from the zero-curvature condition (2.11) that

$$L_{xt_n} - L_{t_n x} = [U_{t_n} - V_{n,x} + [U, V_n], L] = 0.$$

□

Specializing to genus $g = 2$, define

$$(2.18) \quad L(x, t; \lambda) = \frac{1}{4} V(\lambda; Q) + \alpha_1 U(\lambda; Q),$$

$$(2.19) \quad V(\lambda; Q) = 4\lambda^2 U(\lambda; Q) + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

The zero curvature condition for the mKdV equation is

$$U_t - V_x + [U, V] = 0,$$

which yields the scalar modified KdV equation

$$(2.20) \quad u_t + u_{xxx} + 6u^2 u_x = 0, \quad u(x, t) \in \mathbb{R}.$$

The matrices U and V satisfy the following symmetries:

$$(2.21) \quad \begin{aligned} U^\dagger(\lambda^*; Q) &= -U(\lambda; Q), & U^\top(-\lambda; Q) &= -U(\lambda; Q), \\ V^\dagger(\lambda^*; Q) &= -V(\lambda; Q), & V^\top(-\lambda; Q) &= -V(\lambda; Q). \end{aligned}$$

As a consequence the matrix L inherits analogous symmetries:

$$(2.22) \quad L^\dagger(x, t; \lambda^*) = -L(x, t; \lambda), \quad L^\top(x, t; -\lambda) = -L(x, t; \lambda).$$

Proposition 2. *The determinant of L is independent of x and t :*

$$\partial_x \det L = \partial_t \det L = 0.$$

Proof. By Jacobi's formula (or Abel's theorem applying to $L_x = (L_x L^{-1})L$), for any differentiable matrix $L(x, t; \lambda)$

$$\partial_x \det L = \det L \operatorname{tr}(L_x L^{-1}).$$

Using the stationary zero-curvature equation $L_x = [U, L] = UL - LU$ we obtain

$$\partial_x \det L = \det L \operatorname{tr}((UL - LU)L^{-1}) = \det L (\operatorname{tr} U - \operatorname{tr}(L^{-1}LU)) = 0,$$

since $\operatorname{tr}(LUL^{-1}) = \operatorname{tr}(L^{-1}LU)$. The identical argument with $\partial_t L = [V, L]$ yields $\partial_t \det L = 0$. Hence $\det L$ does not depend on x or t . $\square$

Therefore the determinant of L is constant in x and t . Write its factorization as

$$(2.23) \quad \det L(\lambda) = \prod_{j=1}^6 (\lambda - \hat{\lambda}_j).$$

By the symmetry relations (2.22) each root $\hat{\lambda}_j$ forces $\hat{\lambda}_j^*$ and $-\hat{\lambda}_j$ to be roots as well. Two possible root structures follow.

(1) Three purely imaginary pairs:

$$\det L(\lambda) = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2), \quad \lambda_j \in i\mathbb{R}.$$

(2) Nonreal quadruple plus imaginary pair:

$$\det L(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_1^*)(\lambda + \lambda_1)(\lambda + \lambda_1^*)(\lambda - \lambda_2)(\lambda + \lambda_2),$$

with $\lambda_1 \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R})$ and $\lambda_2 \in i\mathbb{R}$.

The matrix elements of L take the explicit form

$$(2.24) \quad \begin{aligned} L_{11} &= -i\lambda^3 - i\lambda\left(\alpha_1 - \frac{1}{2}u^2\right), & L_{22} &= -L_{11}, \\ L_{12} &= u\left(\lambda^2 + i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = u(\lambda - \mu_1)(\lambda - \mu_2), \\ L_{21} &= -u\left(\lambda^2 - i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = -u(\lambda - \mu_1^*)(\lambda - \mu_2^*), \end{aligned}$$

where μ_1, μ_2 denote the (in general complex) roots of the quadratic factor in L_{12}/u . The conjugation in L_{21} is consistent with the reality of u and the symmetry relations (2.22).

Recall that

$$\begin{cases} \Phi_x = U\Phi, \\ \Phi_t = V\Phi, \end{cases} \quad \begin{cases} L_x = [U, L], \\ L_t = [V, L], \end{cases}$$

It follows that $L\Phi$ also satisfies the same Lax pair, since

$$\begin{aligned} (L\Phi)_x &= L_x\Phi + L\Phi_x = (UL - LU)\Phi + LU\Phi = U(L\Phi), \\ (L\Phi)_t &= V(L\Phi). \end{aligned}$$

This observation is also a direct consequence of the identity $L = \Psi = \Phi \sigma_3 \Phi^{-1}$ which implies $L\Phi = \Phi \sigma_3$.

Assume $\Phi(x, t; \lambda)$ is normalized by $\Phi(0, 0; \lambda) = \mathbb{I}$. By uniqueness of solutions of the linear ODEs we obtain

$$(2.25) \quad L(x, t; \lambda) \Phi(x, t; \lambda) = \Phi(x, t; \lambda) L(0, 0; \lambda).$$

In particular, evaluation at $(x, t) = (0, 0)$ yields $L(0, 0; \lambda)$. So knowledge of $L(0, 0; \lambda)$ determines the action of L on the columns of Φ for all (x, t) .

Set

$$(2.26) \quad \det(L(x, t; \lambda)) = \det(L(0, 0; \lambda)) = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = y^2, \quad y = \pm \sqrt{\prod_{j=1}^3 (\lambda^2 - \lambda_j^2)}.$$

A short calculation shows that $\pm iy$ are eigenvalues of $L(0, 0; \lambda)$:

$$\det(\pm iy \mathbb{I} - L(0, 0; \lambda)) = -y^2 - L_{11}^2 - L_{12}L_{21} = -y^2 + \det(L(0, 0; \lambda)) = 0,$$

where L_{ij} denotes the (i, j) -entry of $L(0, 0; \lambda)$.

Hence there exists an invertible matrix M such that

$$(2.27) \quad L(0, 0; \lambda) = M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix} M^{-1}.$$

Combining (2.25) and (2.27) yields

$$L(x, t; \lambda) \Phi(x, t; \lambda) M = \Phi(x, t; \lambda) M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix}.$$

Write the two columns of $\Phi(x, t; \lambda) M$ as

$$\Phi(x, t; \lambda) M = (\Phi_1, \Phi_2),$$

so that

$$L(x, t; \lambda) \Phi_1 = iy \Phi_1, \quad L(x, t; \lambda) \Phi_2 = -iy \Phi_2.$$

Equivalently,

$$(L - iy) \Phi_1 = 0, \quad (L + iy) \Phi_2 = 0,$$

and therefore

$$\ker(L \mp iy) = \text{span}\{\Phi_{1,2}\}.$$

Since each eigenvector may be rescaled, we may write

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \Phi_{11} \begin{pmatrix} 1 \\ r_1 \end{pmatrix}, \quad \Phi_2 = \Phi_{12} \begin{pmatrix} 1 \\ r_2 \end{pmatrix},$$

so

$$\ker(L \mp iy) = \text{span} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix}.$$

Writing the eigenvalue equation explicitly gives

$$\begin{pmatrix} L_{11} \mp iy & L_{12} \\ L_{21} & -L_{11} \mp iy \end{pmatrix} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix} = 0,$$

from which one obtains the two equivalent expressions for the ratio $r_{1,2}$:

$$(2.28) \quad r_{1,2} = \frac{\pm iy - L_{11}}{L_{12}} = \frac{L_{21}}{L_{11} \pm iy}.$$

Denote

$$\mu = \{\mu_1, \mu_2\}, \quad \mu^* = \{\mu_1^*, \mu_2^*\}.$$

The divisor of r_1 is then given by

$$\text{Div}(r_1(x, t; \lambda)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

That is, $r_1(x, t; \lambda)$ vanishes at the points of μ^* , has poles at the points of μ , and has additional normalization determined by the points at infinity $\infty^\pm$. This description of the divisor is central in constructing Baker–Akhiezer functions on the hyperelliptic Riemann surface defined by

$$y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2).$$

Definition 2 (Riemann–Theta function). *Let $\mathcal{R}$ be a compact Riemann surface of genus g . Fix a canonical homology basis and let B denote the associated Riemann period matrix, which is symmetric with negative definite real part. The Riemann–Theta function is the entire function*

$$\Theta(B; z) = \sum_{n \in \mathbb{Z}^g} \exp \left\{ \frac{1}{2} n^\top B n + n^\top z \right\},$$

where

$$n = \begin{pmatrix} n_1 \\ \vdots \\ n_g \end{pmatrix}, \quad n_j \in \mathbb{Z}, \quad z = \begin{pmatrix} z_1 \\ \vdots \\ z_g \end{pmatrix}, \quad z_j \in \mathbb{C}.$$

Proposition 3. *The Riemann–Theta function satisfies:*

(1) *Periodicity in the standard lattice directions:*

$$\Theta(z + 2\pi i e_k) = \Theta(z), \quad k = 1, \dots, g.$$

(2) *Quasi-periodicity in the period matrix directions:*

$$\Theta(z + B e_k) = \Theta(z) e^{-z_k - B_{kk}/2}, \quad k = 1, \dots, g.$$

(3) *Evenness:*

$$\Theta(-z) = \Theta(z).$$

Proof. (1) For $z \mapsto z + 2\pi i e_k$, the exponential in the definition acquires a factor

$$\exp(n^\top (2\pi i e_k)) = \exp(2\pi i n_k) = 1,$$

since $n_k \in \mathbb{Z}$. Thus Θ is periodic in this direction.

(2) For $z \mapsto z + B e_k$, we compute

$$\begin{aligned} \Theta(z + B e_k) &= \sum_{n \in \mathbb{Z}^g} \exp \left\{ \frac{1}{2} n^\top B n + n^\top (z + B e_k) \right\} \\ &= \sum_{n \in \mathbb{Z}^g} \exp \left\{ \frac{1}{2} n^\top B n + n^\top z + n^\top B e_k \right\}. \end{aligned}$$

Write $m = n + e_k$. Then $n = m - e_k$, and the exponent becomes

$$\begin{aligned} &\frac{1}{2} (m - e_k)^\top B (m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k \\ &= \frac{1}{2} (m^\top B m - m^\top B e_k - e_k^\top B m + e_k^\top B e_k) + m^\top z - e_k^\top z + m^\top B e_k - e_k^\top B e_k \\ &= \frac{1}{2} m^\top B m + m^\top z - \frac{1}{2} (m^\top B e_k + e_k^\top B m) + m^\top B e_k - \frac{1}{2} e_k^\top B e_k - e_k^\top z. \end{aligned}$$

Since $B = B^\top$, we have $m^\top B e_k = e_k^\top B m$; moreover, $e_k^\top z = z_k$ and $e_k^\top B e_k = B_{kk}$. Hence

$$\frac{1}{2} (m - e_k)^\top B (m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k = \frac{1}{2} m^\top B m + m^\top z - z_k - \frac{1}{2} B_{kk}.$$

Therefore,

$$\Theta(z + B e_k) = e^{-z_k - B_{kk}/2} \Theta(z).$$

(3) Finally, since the defining series is invariant under $n \mapsto -n$, it follows that

$$\Theta(-z) = \Theta(z).$$

□

Objectives: Construct algebro-geometric mKdV solutions on a genus-two curve by expressing r_1 and u in Riemann theta functions and matching their asymptotics at $\infty^\pm$, poles and zeroes.

The Riemann–Roch theorem provides a fundamental relation between the divisor of a meromorphic function and the corresponding Abel map data on a Riemann surface. In the present setting, for the meromorphic function $r_1(x, t; \lambda)$ we have

$$(3.1) \quad \text{Div}(r_1(x, t; \lambda)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda),$$

which identifies the divisor of r_1 as the difference between two effective divisors supported at the points $P_{\infty^\pm}$ and at the divisor of spectral data μ, μ^* .

A concrete representation of the function r_1 is given in terms of Riemann theta functions:

$$(3.2) \quad r_1(x, t; \lambda) = \hat{r}_1(x, t) \frac{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_\mu))} \exp(\Omega_1(P) - \ln(\omega_1)), \quad P = (\lambda, y).$$

Here Θ denotes the Riemann theta function, $\mathcal{K}$ is the vector of Riemann constants, $\mathcal{A}_{P_0}$ is the Abel map based at P_0 , and α_{P_0} is its linear extension to divisors. The exponential factor is determined by an Abelian integral Ω_1 , while $\hat{r}_1(x, t)$ is a normalizing prefactor independent of λ .

The underlying algebraic curve R_2 associated with this construction is the genus-two hyperelliptic Riemann surface

$$(3.3) \quad R_2 := \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3 + O(\lambda).$$

Thus each point $P \in R_2$ is represented by (λ, y) , where

$$y = \pm \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

To construct the Jacobi variety one chooses a canonical homology basis $\{a_1, a_2; b_1, b_2\}$ on R_2 , satisfying

$$(3.4) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = -b_j \circ a_i = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad i, j = 1, 2.$$

A basis of holomorphic differentials is given by

$$(3.5) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}, \quad d_{ik} \in \mathbb{C},$$

where the normalization guarantees the a -periods form the identity matrix and the b -periods form the Riemann period matrix $B = (B_{ij})$.

The Abel map is defined using these normalized differentials. First, introduce the period lattice

$$(3.6) \quad L_2 = \{z \in \mathbb{C}^2 \mid z = 2\pi i(n + Bm), \quad n, m \in \mathbb{Z}^2\},$$

where B is the period matrix and n, m, z are column vectors in $\mathbb{C}^2$. The Jacobian of R_2 is then

$$(3.7) \quad J(R_2) = \mathbb{C}^2 / L_2.$$

For a base point $P_0 \in R_2$, the Abel map sends a point $P \in R_2$ to

$$(3.8) \quad \mathcal{A}_{P_0} : R_2 \rightarrow J(R_2),$$

$$(3.9) \quad P \mapsto \mathcal{A}_{P_0}(P) = (\mathcal{A}_{P_0,1}(P), \mathcal{A}_{P_0,2}(P)) = \left(\int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

This extends linearly to divisors: for $D = \sum_{P \in R_2} D(P) \cdot P$,

$$(3.10) \quad \alpha_{P_0}(D) = \sum_{P \in R_2} D(P) \mathcal{A}_{P_0}(P).$$

Finally, we introduce the Abelian integral of the second kind,

$$(3.11) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1, \quad \oint_{a_i} d\Omega_1 = 0,$$

with the prescribed asymptotics at the points at infinity,

$$(3.12) \quad \Omega_1(P) = \pm \left(\ln(\lambda) + \ln(\omega_1) + O(1) \right), \quad P \rightarrow \infty^\pm.$$

This integral provides the exponential term appearing in (3.2) and ensures the correct divisor structure of r_1 .

Recall that the function $r_1(x, t; \lambda)$ admits the representation

$$(3.13) \quad r_1(x, t; \lambda) = \frac{L_{21}}{iy + L_{11}},$$

where

$$(3.14) \quad L_{21} = -u \left(\lambda^2 - \frac{i u_x}{2u} \lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2} \right), \quad L_{11} = -i\lambda^3 - i\lambda \left(\alpha_1 - \frac{u^2}{2} \right).$$

To analyze its asymptotic behavior, note first that on the hyperelliptic curve R_2 ,

$$(3.15) \quad y = \pm \left(\lambda^3 - \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda \right) + O(\lambda^{-1}), \quad P \rightarrow \infty^\pm.$$

As $P \rightarrow \infty^-$, the expansion of the denominator yields

$$(3.16) \quad L_{11} + iy = -2i\lambda^3 - i\lambda \left(\alpha_1 - \frac{u^2}{2} \right) + \frac{i}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda + O(\lambda^{-1}),$$

so that

$$(3.17) \quad r_1(x, t; \lambda) = \frac{L_{21}}{iy + L_{11}} = -\frac{i u}{2\lambda} + O(\lambda^{-2}), \quad P \rightarrow \infty^-.$$

Similarly, near ∞^+ one obtains

$$(3.18) \quad r_1(x, t; \lambda) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u}\lambda + O(1), \quad P \rightarrow \infty^+.$$

On the other hand, from the Riemann theta function representation (3.2), the asymptotics at the two points at infinity are

$$(3.19) \quad r_1(x, t; \lambda) = \begin{cases} \hat{r}_1(x, t) \frac{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))} \lambda + O(1), & P \rightarrow \infty^+, \\ \hat{r}_1(x, t) \frac{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{w_1^2 \Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} \lambda^{-1} + O(\lambda^{-2}), & P \rightarrow \infty^-. \end{cases}$$

By comparing these expansions with the direct computation, one obtains the following theta-function formula for the potential $u(x, t)$:

$$(3.20) \quad u = 2i \frac{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{w_1^2 \Theta(\mathcal{K} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

This identity expresses the potential u in terms of theta functions associated with the Jacobian of the spectral curve, thereby linking the algebro-geometric spectral data with the physical field variable.

Recall that the column vector

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}$$

satisfies both parts of the Lax pair:

$$(3.21) \quad \partial_x \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}, \quad \partial_t \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & -V_{11} \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}.$$

Hence the scalar component Φ_{11} satisfies

$$(3.22) \quad \Phi_{11,x} = (-i\lambda + u r_1) \Phi_{11}, \quad \Phi_{11,t} = (V_{11} + V_{12} r_1) \Phi_{11}.$$

Integrating the differential 1-form in (3.22) from a base point (x_0, t_0) to (x, t) yields

$$(3.23) \quad \Phi_{11}(x, t; \lambda) = \Phi_{11}(x_0, t_0; \lambda) \exp \left(\int_{(x_0, t_0)}^{(x, t)} (-i\lambda + u r_1) dx + (V_{11} + V_{12} r_1) dt \right).$$

Thus $\Phi_{11} \in \mathcal{M}(R_2 \setminus \{\infty^\pm\})$, the space of meromorphic functions on R_2 away from the points $\infty^\pm$, and (3.23) shows that Φ_{11} has an essential singularity at $\infty^\pm$.

For large λ , the asymptotic behavior depends on the sheet. As $\lambda \rightarrow \infty$ with $P \rightarrow \infty^-$,

$$(3.24) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \exp(-i\lambda(x + 4\lambda^2 t)) + O(1), \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = (0 + O(\lambda^{-1})) \exp(-i\lambda(x + 4\lambda^2 t)). \end{aligned}$$

On the other sheet, the expansion of r_1 near ∞^+ is

$$(3.25) \quad r_1(x, t; \lambda) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u} \lambda + \frac{u_x}{u^2} + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Consequently,

$$(3.26) \quad -i\lambda + u r_1 = i\lambda + (\ln u)_x + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Substitution into (3.23) gives the asymptotics

$$(3.27) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &\rightarrow u(x, t) \exp(i\lambda(x + 4\lambda^2 t)), \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} \rightarrow 2i\lambda \exp(i\lambda(x + 4\lambda^2 t)). \end{aligned}$$

Hence, as $P \rightarrow \infty^+$,

$$(3.28) \quad \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \Phi_{1,0}^+(x_0, t_0) \left(\begin{pmatrix} 0 \\ 2i\lambda \end{pmatrix} + O(1) \right) \exp(i\lambda(x + 4\lambda^2 t)),$$

where the normalization constant is

$$(3.29) \quad \Phi_{1,0}^+ = \lim_{P \rightarrow \infty^+} \Phi_{11}(x_0, t_0; \lambda).$$

Recall

$$(3.30) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)},$$

$$(3.31) \quad r_1(\lambda) = \frac{L_{21}(\lambda)}{iy(\lambda) + L_{11}(\lambda)} = \frac{-u(\lambda - \mu_1^*)(\lambda - \mu_2^*)}{iy + L_{11}}.$$

We write

$$L(\lambda) = \begin{pmatrix} L_{11}(\lambda) & L_{12}(\lambda) \\ L_{21}(\lambda) & -L_{11}(\lambda) \end{pmatrix}, \quad U(\lambda) = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix},$$

and recall the stationary zero-curvature relation

$$(3.32) \quad L_x = [U, L].$$

With $\sigma_3 = \text{diag}(1, -1)$ and off-diagonal part $L^{\text{off}} = \begin{pmatrix} 0 & L_{12} \\ L_{21} & 0 \end{pmatrix}$, we compute

$$\begin{aligned}
(3.33) \quad [U, L] &= -2i\lambda \sigma_3 L^{\text{off}} + [Q, L] \\
&= -2i\lambda \sigma_3 L^{\text{off}} + \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} - \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \\
&= -2i\lambda \begin{pmatrix} 0 & L_{12} \\ -L_{21} & 0 \end{pmatrix} + \begin{pmatrix} u(L_{21} + L_{12}) & -2uL_{11} \\ -2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix} \\
&= \begin{pmatrix} u(L_{21} + L_{12}) & -2i\lambda L_{12} - 2uL_{11} \\ 2i\lambda L_{21} - 2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix}.
\end{aligned}$$

Thus from (3.32) we obtain

$$\begin{aligned}
(3.34) \quad \partial_x L_{11} &= u(L_{21} + L_{12}), \\
\partial_x L_{12} &= -2i\lambda L_{12} - 2uL_{11}, \\
\partial_x L_{21} &= 2i\lambda L_{21} - 2uL_{11}.
\end{aligned}$$

By assumption

$$(3.35) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2).$$

Differentiating with respect to x gives

$$(3.36) \quad \partial_x L_{12}(\lambda) = u_x(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,x}(\lambda - \mu_2) - u \mu_{2,x}(\lambda - \mu_1).$$

Evaluating (3.36) at $\lambda = \mu_1$ leaves only the middle term:

$$(3.37) \quad \partial_x L_{12}(\mu_1) = -u \mu_{1,x}(\mu_1 - \mu_2).$$

On the other hand, from (3.34) evaluated at $\lambda = \mu_1$ we have

$$(3.38) \quad \partial_x L_{12}(\mu_1) = -2u L_{11}(\mu_1).$$

Equating the two expressions yields

$$(3.39) \quad \mu_{1,x} = \frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2}.$$

Next, the determinant relation $y^2 = \det L = -L_{11}^2 - L_{12}L_{21}$ at $\lambda = \mu_1$ reduces to

$$(3.40) \quad y(\mu_1)^2 = -L_{11}(\mu_1)^2,$$

since $L_{12}(\mu_1) = 0$. Hence

$$(3.41) \quad -iy(\mu_1) = \pm L_{11}(\mu_1).$$

We adopt the choice

$$(3.42) \quad -iy(\mu_1) = L_{11}(\mu_1).$$

Consider now the principal part of $u r_1$ near $\lambda = \mu_1$. From (3.30) and (3.35),

$$(3.43) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_1) - L_{11}(\mu_1)}{(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using (3.42) and (3.39),

$$iy(\mu_1) - L_{11}(\mu_1) = -2L_{11}(\mu_1) = -(\mu_1 - \mu_2) \mu_{1,x}.$$

Thus (3.43) becomes

$$(3.44) \quad u r_1(\lambda) = -\frac{\mu_{1,x}}{\lambda - \mu_1} + O(1) = \partial_x(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

Then by (3.23) and (3.44) we conclude that $\lambda = \mu_1$ is a simple zero of $\Phi_{11}(x, t; \lambda)$.

Supplementary computations:

Evaluate (3.36) at $\lambda = \mu_2$. Only the third term survives, so

$$(3.45) \quad \partial_x L_{12}(\mu_2) = -u \mu_{2,x}(\mu_2 - \mu_1).$$

From (3.34) at $\lambda = \mu_2$ we have

$$(3.46) \quad \partial_x L_{12}(\mu_2) = -2u L_{11}(\mu_2).$$

Equating these expressions yields

$$(3.47) \quad \mu_{2,x} = \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1}.$$

The determinant identity $y^2 = \det L = -L_{11}^2 - L_{12}L_{21}$ at $\lambda = \mu_2$ gives, since $L_{12}(\mu_2) = 0$,

$$(3.48) \quad y(\mu_2)^2 = -L_{11}(\mu_2)^2,$$

and therefore

$$(3.49) \quad -iy(\mu_2) = \pm L_{11}(\mu_2).$$

Adopting the same sheet convention as for μ_1 we take

$$(3.50) \quad -iy(\mu_2) = L_{11}(\mu_2).$$

Compute the principal part of $u r_1$ near $\lambda = \mu_2$. From (3.30) and (3.35),

$$(3.51) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_2) - L_{11}(\mu_2)}{(\mu_2 - \mu_1)} \cdot \frac{1}{\lambda - \mu_2} + O(1), \quad \lambda \rightarrow \mu_2.$$

Using (3.50) and (3.47) we obtain

$$iy(\mu_2) - L_{11}(\mu_2) = -2L_{11}(\mu_2) = -(\mu_2 - \mu_1) \mu_{2,x}.$$

Hence (3.51) becomes

$$(3.52) \quad u r_1(\lambda) = -\frac{\mu_{2,x}}{\lambda - \mu_2} + O(1) = \partial_x(\ln(\lambda - \mu_2)) + O(1), \quad \lambda \rightarrow \mu_2.$$

Therefore, by (3.23) and (3.52), $\lambda = \mu_2$ is a simple zero of $\Phi_{11}(x, t; \lambda)$.

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